Continuity and spatial averaging in ion-acoustic reconnection diagnostics

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Abstract

Li, Liu and Loureiro investigate ion-acoustic instability and ion heating in collisionless magnetic reconnection. We examine two derivational issues in the final journal article, separately from its observed waves and temperature histories. A temperature estimate in Appendix B divides by the in-plane ion velocity divergence. Steady continuity at a two-dimensional stagnation point with positive density sets that divisor to zero. Eliminating the normalized moment subsystem without division instead gives a compatibility condition, leaving the out-of-plane temperature undetermined when the corresponding heat-flux divergence vanishes. Explicit positive-temperature subsystem witnesses illustrate the missing closure; they are not complete kinetic solutions. A second issue concerns spatial averaging. For a finite Savitzky-Golay filter, the covariance of two fields generally differs from the filtered product of their pointwise residuals. The source's drag and time-dependent fluctuation formulas therefore require an explicit residual convention. Subtracting the same center mean throughout each window supplies a valid alternative and exactly recovers the covariance. Synthetic rational examples and independent controls demonstrate both outcomes. The original postprocessing implementation and underlying fields were not obtained, so no error in the plotted anomalous terms or revised reconnection rate is established. The supported revision concerns a singular temperature derivation and reproducible operator definitions, rather than a refutation of the reported instability or heating.

1. Source, version and scope

Ion-acoustic waves can redistribute momentum and energy in a reconnecting plasma without necessarily controlling its global reconnection rate. Li, Liu and Loureiro [1,2] study this distinction using two-dimensional kinetic simulations with several initial ion-to-electron temperature ratios. They report instability in cold-ion cases, substantial ion heating, localized anomalous momentum terms and comparatively little change in the overall reconnection rate. A consistency check of their analytical diagnostics must distinguish these observations from the reduced arguments offered to explain them.

Our principal source is the 30-page version of record, *Journal of Plasma Physics* **92**(1), E20 (2026), DOI 10.1017/S002237782510113X [1], published online 27 February 2026. The supplied 28-page preprint is byte-identical to the official arXiv:2505.08983v1 PDF [2]. Both editions were read completely, including captions, footnotes, appendices and references. Their SHA-256 identifiers are respectively `0ad1cbf7 95a58958b6889bebdff355fd463fb343aadef84e12e9261e100d7fa0` and `ec0dab848877c079b103cb9f 22f5efafddda3b7576a4955695bb12b19f2ffd60`. Equation numbers and pages below refer to [1]. A local institutional copy adds a report cover; it is not treated as an independent scientific edition.

The final article replaces the preprint's MHD-compressional-mode comparison with a uniform-flow Doppler shift, $\omega_{\mathrm{DS}} = \omega_{\mathrm{Vlasov}} \pm k u_{x,i,\max}$, in Eq. (4.1) and revised Fig. 6. It also justifies the initial drifting-Maxwellian comparison, qualifies its longitudinal and three-dimensional applicability, and explicitly labels its physical-mass-ratio extrapolation speculative. We credit these changes. The su-

perseded preprint overlays are not alleged as errors in the journal article. The anomalous terms are renumbered from preprint Eqs. (4.6)-(4.9) to journal Eqs. (4.2)-(4.5).

The narrower questions are whether the Appendix B temperature reduction is determined under its own continuity assumptions, and what averaging convention makes Appendix D's covariance terms exact. The inspected arXiv source archive contains manuscript/support files and eleven figure PDFs, but no case-specific numerical analysis program, input decks or underlying particle/field arrays. The journal says supporting data are available upon reasonable request; no source-author correspondence was obtained for this study. Bounded official-record and correction searches on 29 October 2026 identified no target-specific correction. This is not proof of its absence, and an erratum for a different ion-acoustic-turbulence paper is excluded.

2. Continuity before division

Let $d = \partial_x u_{ix} + \partial_y u_{iy}$ at the two-dimensional X-point. The exact continuity equation is

$$\partial_t n_i + \boldsymbol{u}_{i\perp} \cdot \nabla_\perp n_i + n_i d = 0. \tag{1}$$

The source's Appendix B assumes a steady, uniformly distributed plasma near that point and sets the in-plane ion velocity there to zero. With $n_{i,X} > 0$, Eq. (1) gives $d = 0$. Uniform density is stronger than needed: at an exactly steady stagnation point, the advective term already vanishes. The source explicitly uses incompressibility again in Eq. (B13), p. 21, $\partial_x u_{ix}|_X = -\partial_y u_{iy}|_X$.

Nevertheless, its rearranged Eq. (B6c), p. 20, is

$$T_{i,zz,X} = m_i u_{iz,X}^2 - \frac{1}{n_{i,X}} \frac{\partial_x q_{i,zzx}|_X + \partial_y q_{i,zzy}|_X}{\partial_x u_{ix}|_X + \partial_y u_{iy}|_X}. \tag{2}$$

This divides by precisely d. Setting the numerator's heat-flux divergence to zero does not make a zero divisor legitimate. The subsequent estimate in Eq. (B10b), which additionally assumes order-unity plasma beta as the source notes,

$$T_{i,zz,X} \sim \frac{2(T_{e0} + 2T_{i0})^2}{T_{e0} + T_{i0}} \sim 2T_{e0} \quad (T_{i0} \ll T_{e0}), \tag{3}$$

therefore is not fixed by this divided relation under the stated exact steady assumptions. This concerns its derivation, not a claim that a measured value near this scale is impossible.

A direct normalized check avoids silently repairing the source's density conventions. In printed (B5c), the off-diagonal temperature-gradient cross term lacks an additional density multiplier, while (B5d) uses that gradient sum directly. We consequently restrict the following *literal subsystem* elimination to $n_i = m_i = e = 1$. Write $u = u_{iz,X}$, $T = T_{i,zz,X}$, $S = \partial_x T_{i,xz}|_X + \partial_y T_{i,yz}|_X$, and $Q = \partial_x q_{i,zzx}|_X + \partial_y q_{i,zzy}|_X$. Equations (B5c)-(B5d) then read

$$Q + (T + u^2)d + 2uS = 2E_z u, \qquad E_z = S + ud. \tag{4}$$

Subtracting twice u times the second equation gives the undivided condition

$$Q + (T - u^2)d = 0. \tag{5}$$

At $d = 0$, it requires $Q = 0$, while the second equation fixes $E_z = S$. Neither selects T. For example, $u = 2$, $E_z = S = 3$, $Q = d = 0$ satisfy both displayed equations for $T = 1$ and for $T = 7$. Their common squared bulk speed $u^2 = 4$ does not determine their thermal variance. These exact positive-temperature witnesses demonstrate underdetermination of this moment subsystem. They do not supply a complete distribution satisfying the full moment hierarchy or the Vlasov-Maxwell system.

An approximately steady argument could retain a small nonzero d. It would then need consistent relative orders: if $d = \epsilon d_1$ and $Q = \epsilon Q_1$, Eq. (5) gives $T = u^2 - Q_1/d_1$, even as both tend to zero. Dropping Q before resolving its ratio to d imposes an additional closure. Nonzero-divergence compatibility paths $S = 3 - 2d$, $Q = (4 - T)d$ make this explicit. Their finite-d points intentionally violate exact steady uniform-density continuity; their seven selected positive-temperature endpoints share $Q = d = 0$. They are not alternative steady reconnection solutions.

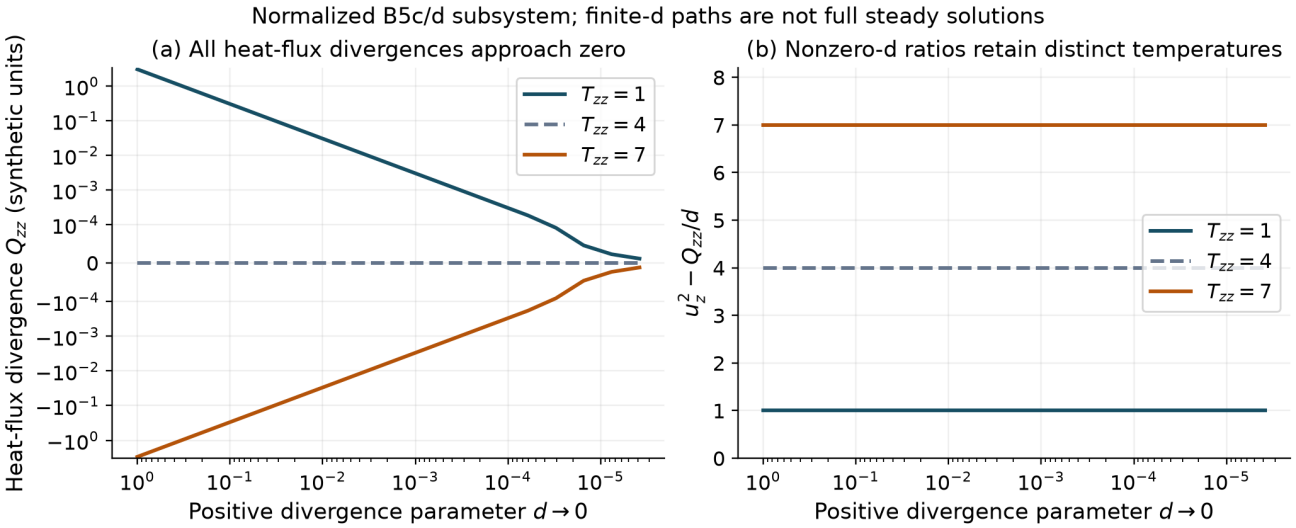


Figure 1. With $n_i = m_i = e = 1$, $u = 2$ and $E_z = 3$, each selected positive T has $S = 3 - 2d$ and $Q = (4 - T)d$. All paths satisfy the two normalized equations exactly. At $d = 0$, both numerator and divisor of the rearranged expression vanish, and the temperature remains a parameter. Finite-d paths fail the separately stated steady uniform-density continuity assumption.

The source already recognizes a failure of its zero-heat-flux approximation for the xx and yy moments on p. 21. That acknowledgment should be retained. It does not remove the zero divisor in the zz rearrangement. A related step, Eq. (B17), p. 22, differentiates and extends a relation specialized to the X-point to the sheet edge, where the preceding equations retain advection and magnetic terms absent at the X-point. A neighborhood closure is needed to infer that edge derivative and the range in (B18). Neither issue refutes the temperatures plotted in Figs. 8-9. An electric field may contribute to heating through particle orbits and phase-space mixing; a bulk acceleration alone does not determine a central second moment.

3. What a finite spatial mean implies

Appendix D, pp. 23-24, decomposes each quantity as $f = \langle f \rangle + \delta f$ and specifies two-dimensional spatial Savitzky-Golay averaging. Its rationale separates small ion-acoustic structures from larger reconnection structure. Fig. 10 labels approximately $0.22\, d_{e0}^{-1}$ as the reciprocal window length. The exact point-count window, polynomial degree, two-dimensional construction, edge mode and residual implementation were not obtained. Window length alone does not specify a Savitzky-Golay frequency response [3].

For clarity, let a fixed normalized linear filter be

$$(Gf)(\boldsymbol{x}) = \sum_{\boldsymbol{r}} w_{\boldsymbol{r}} f(\boldsymbol{x}+\boldsymbol{r}), \qquad \sum_{\boldsymbol{r}} w_{\boldsymbol{r}} = 1. \tag{6}$$

Write $N = Gn$, $F = Gf$, and first interpret the fluctuation as the pointwise field $f' = f - Gf$. The exact covariance and filtered residual product are distinct objects:

$$C_{nf} = G(nf) - NF, \qquad R_{nf} = G(n'f'). \tag{7}$$

Expanding without Reynolds factorization gives

$$\begin{aligned} C_{nf} - R_{nf} = G[(Gn)(Gf)] - (Gn)(Gf) \\ + G[(Gn)f'] + G[n'(Gf)] \equiv K_{nf}. \end{aligned} \tag{8}$$

Normalization and linearity do not set these extra terms to zero. In particular, $Gf' = Gf - G^2 f$ need not vanish. The distinction between general spatial filters and Reynolds averages is established filtering theory [4], rather than a newly discovered failure of averaging. The source-specific question is which of these operations its formulas and postprocessing implement.

For species mass m, signed charge q, and $U = G\boldsymbol{u}$, source Eqs. (4.2) and (4.5) use residual-product expressions for drag and the time-dependent fluctuation term. Under the pointwise interpretation, an exact covariance formulation is instead

$$\boldsymbol{D}_* = -\frac{C_{n\boldsymbol{E}}}{N}, \qquad \boldsymbol{V}_* = \frac{m}{qN}\,\partial_t C_{n\boldsymbol{u}}. \tag{9}$$

Here the filter is fixed in time and N remains outside the time derivative. Replacing C by R changes these terms by quantities determined by Eq. (8); it is not justified by the field decomposition alone. Conversely, the source's $\boldsymbol{T}$ and $\boldsymbol{I}$ already retain full products:

$$\begin{aligned} \boldsymbol{T} &= -\frac{G[n(\boldsymbol{u}\times\boldsymbol{B})]}{Nc} + \frac{(G\boldsymbol{u})\times(G\boldsymbol{B})}{c}, \\ \boldsymbol{I} &= \frac{m}{qN}\left\{G[\nabla\cdot(n\boldsymbol{u}\boldsymbol{u})] - \nabla\cdot(NUU)\right\}. \end{aligned} \tag{10}$$

They are positive controls for this particular algebraic issue. Derivative commutation is a separate question: a constant interior convolution commutes with the corresponding translation-invariant derivative, while a position-dependent boundary rule need not. An illustrative edge commutator does not establish the source's unspecified edge treatment is wrong.

4. An exact counterexample and an exact alternative

The five-point, degree-two Savitzky-Golay weights [3] are $(-3, 12, 17, 12, -3)/35$ at offsets $r = -2, -1, 0, 1, 2$. This kernel reproduces quadratics. Set $n(x) = 2 + x^2$ and $E(x) = x^2$. Then $n' = E' = 0$ everywhere for the infinite polynomial fields, so $R_{nE} = 0$. At $x = 0$, however,

$$N = 2, \quad GE = 0, \quad C_{nE} = -\frac{72}{35}, \quad D_* = \frac{36}{35}, \quad D_{\text{residual}} = 0. \tag{11}$$

The nonzero value comes from the degree-four product, which the degree-two kernel does not reproduce. Negative Savitzky-Golay weights allow a negative weighted covariance; this is not a negative probabilistic variance or negative physical density. Both the density and its filtered central value are positive. A separable two-dimensional filter acting on fields independent of y gives the same example, so the algebra is not rescued merely by specifying two dimensions.

A weaker density variation, $n = 2 + x^2/100$, yields $C_{nE} = -18/875$ and $D_* = 9/875$. It can be embedded in an exact static primitive force balance with positive pressure on the stencil neighborhood: take $q = 1$, $\boldsymbol{u} = \boldsymbol{B} = 0$ and $P(x) = 100 + (2/3)x^3 + x^5/500$, so $\partial_x P = nE$. Filtering that balance requires the full product $G(nE)$. This is a deliberately constructed moment-level counterexample, not a proposed reconnection equilibrium or a replacement for any source field.

There is an essential counterargument to a source-code error allegation. One can instead define a window-centered residual,

$$\delta_{\boldsymbol{x}} f(\boldsymbol{r}) = f(\boldsymbol{x} + \boldsymbol{r}) - (Gf)(\boldsymbol{x}). \tag{12}$$

The same center mean is then subtracted at every sample in that evaluation window. Direct expansion and normalization give

$$\sum_{\boldsymbol{r}} w_{\boldsymbol{r}} \delta_{\boldsymbol{x}} n(\boldsymbol{r}) \delta_{\boldsymbol{x}} f(\boldsymbol{r}) = G(nf)(\boldsymbol{x}) - (Gn)(\boldsymbol{x})(Gf)(\boldsymbol{x}). \tag{13}$$

This is exact even for signed weights. It reproduces the nonzero covariance in Eq. (11), while the pointwise residual field gives zero. If the source implementation uses Eq. (12), its residual products can already be the correct covariances. The available definition and absent implementation do not decide between these conventions. Thus the defensible source finding is an operator-definition and reproduction gap, not a demonstrated error in Fig. 10 amplitudes.

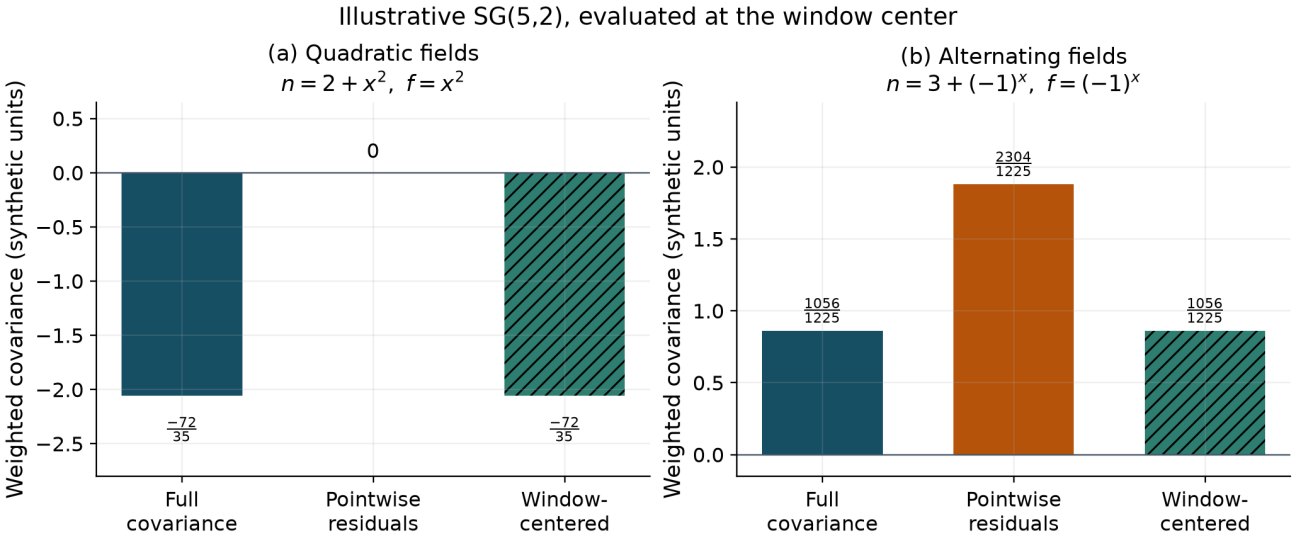


Figure 2. Two original dimensionless examples use a five-point, degree-two kernel. The polynomial case and an alternating-field case distinguish full covariance from the filtered pointwise residual product. The window-centered calculation agrees exactly with full covariance in both. Neither example reconstructs the original filter, density, electric field or plotted diagnostic.

The appropriate numerical repair, if needed, is to define the filter and residual evaluation explicitly, compare the full covariance with the implemented fluctuation product, and close the filtered momentum budget using the same discrete derivatives. Without the original arrays, the magnitude, sign and

dynamical importance of any omitted term remain unknown. A local force fluctuation also is not by itself a global reconnection-rate or irreversible-heating measurement; the source separately presents direct rate and temperature histories.

5. Verification and retained controls

Original calculations [5] construct four synthetic centered Savitzky-Golay kernels using exact rational least squares: window/degree pairs $(5, 2)$, $(7, 3)$, $(9, 2)$ and $(5, 4)$. They check polynomial reproduction, covariance decomposition and the exact window-centered alternative in 20 cases. Forty time-dependent controls check $mC_{nf}/(qN)$ with both charge signs, including an instantaneous force balance with $\boldsymbol{u} = \boldsymbol{E} = \boldsymbol{B} = 0$, $\partial_t \boldsymbol{u} = f$ and $\nabla \cdot P = -mnf$. The corresponding primitive momentum equation is satisfied at that instant; no full kinetic evolution is asserted. Constants, the degree-four identity kernel, a true broadcast ensemble mean, and the source's full-product terms are retained controls.

The moment producer checks 273 exact states: seven positive selected temperatures, each with $d = \pm 2^{-k}$ for $k = 0, \ldots, 18$ and $d = 0$. All satisfy the literal normalized (B5c)-(B5d) subsystem. The 266 nonzero-divergence states recover their selected temperature only when division is legitimate, and all seven zero-divergence cases mark Eq. (2) undefined. The $T = 4$ family is the zero-heat-flux, nonzero-divisor positive control. Continuity violations at every finite-d point are explicitly reported, rather than presented as physical steady states.

Separate independent programs do not import the producer. One checks filter identities, signed-charge time-dependent terms, derivative commutation and the window-centered alternative using exact fractions. Another verifies Maxwellian response identities, Harris/grid normalization, homogeneous Doppler shifts and exact finite-mass marginal thresholds, as well as two normalized moment witnesses. For mass ratio 100, its cold-ion threshold controls give $U_d/c_s \simeq 0.681520$ and 1.354021, consistent with the source's stated 0.68 and 1.36 under the declared local normalization. The Maxwellian susceptibility and the final uniform nonrelativistic Doppler transformation pass; no dispersion-calculation error is asserted from these controls.

An initially considered Appendix E mass-factor criticism is withdrawn. Inverting its scale inequality for the existence of *some* scale below d_i differs from requiring the condition at *every* scale above d_e. The latter uses $d_e/d_i = \sqrt{m_e/m_i}$ and recovers the printed mass factor in (E3). Its wording admits that universal interpretation. Neither that candidate, a typical wavelength/FWHM comparison, nor acknowledged mildly relativistic velocities establishes an error in the PIC data. The source's explicit finite-mass-ratio and three-dimensional caveats remain relevant.

The plotting producer was run under CPython 3.12.14, NumPy 2.3.5 and Matplotlib 3.11.2; its algebra uses exact fractions, with floating point restricted to plotting. Independent filter checks use the standard library under CPython 3.13.7. Independent response/marginal checks use CPython 3.13.7, NumPy 2.2.1 and SciPy 1.17.1. Actual environments and original receipts are retained separately. Equality in these controls is internal algebraic and computational evidence, not independent human peer review or guaranteed bitwise portability across environments.

6. Consequences and limits

The temperature conclusion supported here is precise: under the stated exact steady continuity assumptions, the divisor in (B6c) vanishes, and its zero-heat-flux use does not determine (B10b). A consistent approximate treatment must supply the missing relative order or a different closure. This does not establish that the measured out-of-plane temperature is incorrect, that the entire hierarchy has an alternative steady solution, or that electric fields cannot participate in heating.

For the anomalous terms, full covariances supply an exact operator formulation, but the window-centered interpretation already yields them. The source's actual code, fields and filtered budgets would be needed to decide whether any numerical correction is required. No revised anomalous amplitude, reconnection rate, physical mass-ratio simulation, entropy-production result or complete kinetic reproduction is supplied. The final edition's observed instability, measured temperature histories and reported rate comparison remain separate evidence.

A focused revision could remove the zero-divisor inference, state the assumptions needed for any replacement temperature estimate, and document the exact filter kernel and residual convention with an explicit budget-closure check. These steps make the analytical interpretation reproducible while preserving conclusions that the present audit has not tested.

Code and data availability. Original code, rational outputs, two original figures, source-edition provenance and portable independent checks accompany this study [5]. The verified software archive is `https://doi.org/10.5281/zenodo.23782239` , direct record `https://zenodo.org/records/23782239` , frozen at computational commit `36b1c070e029fd7e3c05e96af3b93172444bfee2` in `https://github.com/Teylersf/ion-acoustic-moment-filter-audit` . All three programs passed from a clean 42-file exported tree without changing any archived file bytes. The anonymous archive download matches that snapshot: 363,538 bytes, SHA-256 `33d3f51c355896403154b1be4f4e13a2a652f69e80628ac9c8512abc97abd7b3`. The DOI resolver and primary DataCite registry were checked on 29 October 2026; its state is `findable`. The archive records actual environments and distinguishes normalized subsystem tests from unavailable original simulation data. Source PDFs and original figure files are not presented as this author's work.

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